

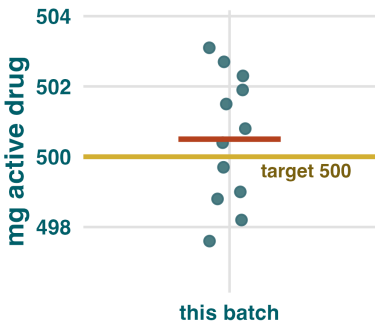
The t-test family | each type, its hypotheses, and its direction

ONE-SAMPLE

One group's mean vs a **fixed target**.
e.g. is a batch's mean dose really the labelled 500 mg?



?
really 500?



Gold line: 500 mg target. Red bar: batch mean.

```
pill <- c(498.2, 501.5, 499.7, 502.3, 500.8,
         497.6, 503.1, 499.0, 501.9, 500.4,
         498.8, 502.7)
t.test(pill, mu = 500) # alpha = 0.05
#> t = 0.94, df = 11, p-value = 0.368
```

R

HYPOTHESES & DIRECTION

$H_0 : \mu = 500$

two-sided $H_1 : \mu \neq 500$ `t.test(pill, mu = 500, alternative = "two.sided")`

one-sided $H_1 : \mu > 500$ `t.test(pill, mu = 500, alternative = "greater")`

one-sided $H_1 : \mu < 500$ `t.test(pill, mu = 500, alternative = "less")`

Here: two-sided. Over- or under-dosing both fail, so watch both tails. The cautious default.

A nurse compares the resting heart rate of 20 athletes with the adult norm of 72 bpm, looking for any difference. Which alternative hypothesis fits?

✓ C

A. $\mu > 72$

B. $\mu < 72$

C. $\mu \neq 72$

For the athlete heart-rate test against the adult norm of 72 bpm, what is the null hypothesis?

✓ B

A. $\mu > 72$

B. $\mu = 72$

C. $\mu < 72$

Her test returns $p = 0.02$ at $\alpha = 0.05$. What should she conclude?

✓ A

A. Reject H_0

B. Fail to reject H_0

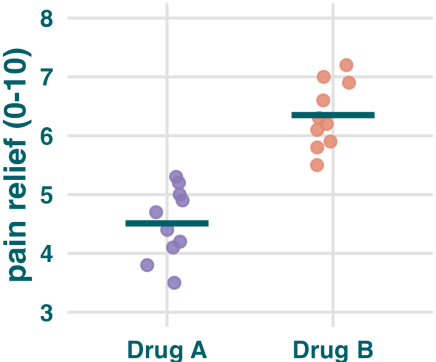
C. H_0 is proven

TWO-SAMPLE (INDEPENDENT)

Means of **two separate groups**.
e.g. does painkiller A or B give more relief?



vs
more relief?



Two independent clouds. Teal bars: group means.

```
A <- c(4.1, 5.2, 3.8, 4.9, 4.4,
       5.0, 3.5, 4.7, 4.2, 5.3)
B <- c(6.3, 5.8, 6.9, 6.1, 7.2,
       5.5, 6.6, 7.0, 6.2, 5.9)
t.test(B, A) # alpha = 0.05
#> t = 7.04, df = 17.9, p-value < 0.001
#> p < 0.05 -> reject H0
```

R

HYPOTHESES & DIRECTION

$H_0 : \mu_A = \mu_B$

two-sided $H_1 : \mu_A \neq \mu_B$ `t.test(B, A, alternative = "two.sided")`

one-sided $H_1 : \mu_B > \mu_A$ `t.test(B, A, alternative = "greater")`

one-sided $H_1 : \mu_B < \mu_A$ `t.test(B, A, alternative = "less")`

Either can apply. Use one-sided only if "is B better?" is the whole question, decided **before** the data.

Researchers compare mean cholesterol in two separate groups: one on a Mediterranean diet, one on their usual diet. What is the null hypothesis?

✓ A

A. $\mu_{\text{med}} = \mu_{\text{usual}}$

B. $\mu_{\text{med}} < \mu_{\text{usual}}$

C. $\mu_{\text{med}} \neq \mu_{\text{usual}}$

Why are two-sided t-tests often preferred in clinical drug testing?

✓ B

A. They make it easier to claim a drug works by lowering the p-value.

B. They detect either improvement or underperformance, and reduce directional bias.

PAIRED (DEPENDENT)

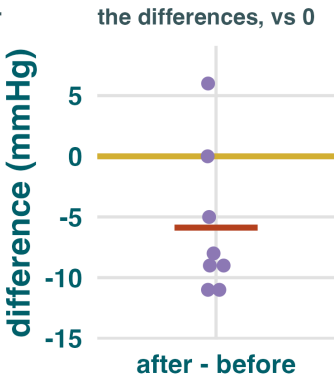
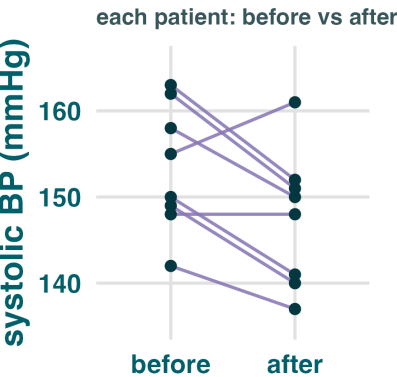
The **same group** measured **twice**.
e.g. does BP fall from before to after a drug?



before
155

6 wks

after
147



Each before→after pair becomes one difference, tested against 0.

```
before <- c(150,148,162,142,155,149,163,158)
after <- c(141,148,151,137,161,140,152,150)
t.test(after, before, paired = TRUE) # a=0.05
t.test(after - before, mu = 0) # same
#> t = -2.76, df = 7, p-value = 0.028
#> p < 0.05 -> reject H0
```

R

HYPOTHESES & DIRECTION ($d = \text{after} - \text{before}$)

$H_0 : \mu_d = 0$

two-sided $H_1 : \mu_d \neq 0$ `t.test(after, before, paired = TRUE, alternative = "two.sided")`

one-sided $H_1 : \mu_d < 0$ `t.test(after, before, paired = TRUE, alternative = "less")`

one-sided $H_1 : \mu_d > 0$ `t.test(after, before, paired = TRUE, alternative = "greater")`

For each study, name the t-test and a fitting H_1 :

1. Same runners timed before and after an 8-week program, hoping for faster times.

✓ paired, $H_1 : \mu_d < 0$

2. Mean recovery time for two surgical methods, on two separate groups of patients.

✓ two-sample, $H_1 : \mu_1 \neq \mu_2$

3. Mean volume in a sample of vials, checked against the labelled 0.5 mL.

✓ one-sample, $H_1 : \mu \neq 0.5$